Resistance

Exam code: 0625 & 0972

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Ohm's law

Resistance

- **Resistance** is defined as:

The opposition to current

- Resistance occurs because the free **electrons** flowing in the circuit (current) **collide** with the metal ions in the wire
- These collisions **slow down** the electrons, or, in other words, **resist** their flow
  - The **higher** the resistance of a circuit, the **lower** the current
  - This means that good conductors have a **low** resistance and insulators have a **high** resistance
- The resistance of a circuit can be **increased** by adding resistors (or variable resistors) to it
- Every electrical component has a resistance, even wires
  - In exam questions, the resistance of the wires and batteries are assumed to be negligible

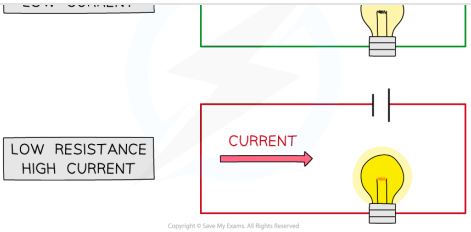
The effect of resistance on the current in a circuit

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



When a circuit has a high resistance, a lower current will flow, and vice versa

Ohm's law

- Current, I, potential difference, V, and resistance, R, all affect one another
- Changing any **one** of these in a circuit, changes **all** of them
- Current and resistance are **inversely proportional**
  - If the resistance is doubled, current will halve
- This relationship is described by the following equation, known as Ohm's law

$$R = \frac{V}{I}$$

- Where
  - R = resistance, measured in ohms (Ω)
  - V = potential difference, measured in volts (V)
  - I = current, measured in amperes or amps (A)

Consequences of Ohm's law

- Resistors are used in circuits to control either:
  - the current in branches of the circuit (through certain components)
  - the potential difference across certain components
- This is due to the consequences of Ohm's Law
  - The current in an electrical conductor decreases as its resistance increases (for a constant p.d.)
  - The p.d. across an electrical conductor increases as its resistance increases (for a constant current)

Worked Example

A 12 Ω resistor has a current of 0.3 A flowing through it.

Determine the potential difference across the resistor.

Answer:

Step 1: List the known quantities

- Resistance, $R = 12\,\Omega$
- Current, $I = 0.3\text{ A}$

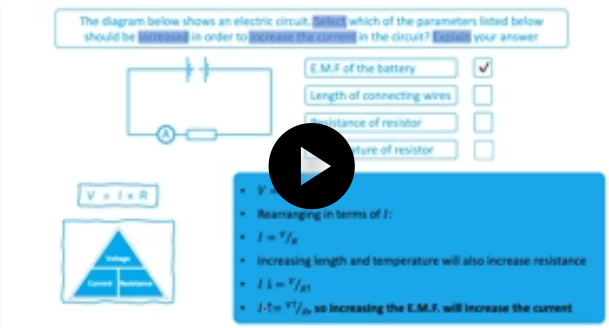
$$R = \frac{V}{I}$$

$$V = IR$$

Step 3: Substitute in the known values to calculate

$$V = 0.3 \times 12$$

$$V = 3.6 \text{ V}$$



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Current-voltage graphs

Extended tier only

- The relationship between current and potential difference of a component can be shown on a current-voltage (*I*-*V*) graph
- When the relationship between current and potential difference is **linear**:
  - the *I*-*V* graph is a **straight line** which passes through the **origin**
  - the resistance is **constant**
  - these are known as **ohmic resistors**
- When the relationship between current and voltage is **non-linear**:
  - the *I*-*V* graph that is **not** a straight line
  - the resistance is **not** constant
  - these are known as **non-ohmic resistors**

Current-voltage (I-V) graph for a resistor and a filament lamp

Linear IV graphs are straight lines through the origin, indicating a constant resistance. Non-linear IV graphs are curved, indicating a variable resistance

- Components with **linear** *I-V* graphs (ohmic resistors) include:
  - fixed resistors (at constant temperature)
  - wires (at constant temperature)
- Components with **non-linear** *I-V* graphs (non-ohmic resistors) include:
  - filament lamps
  - diodes
  - LDRs
  - thermistors

I-V graph for ohmic conductors

- The relationship between current and voltage for a wire or fixed resistor is linear, or **directly proportional**, which means
  - the IV graph is a straight line, so voltage and current increase (or decrease) by the **same** amount
  - the slope of the graph is constant, so resistance is **constant**

I-V graph for a a wire of fixed resistor

The current is directly proportional to the potential difference (voltage) as the graph is a straight line through the origin

I-V graph for a filament lamp

    - the *IV* graph is not a straight line, so voltage and current do **not** increase (or decrease) by the same amount
    - the slope of the graph is not constant, so resistance **changes**

- The *IV* graph for a filament lamp shows as voltage increases
    - the current increases at a proportionally **slower** rate
    - the resistance **increases**; the flatter the slope, the higher the resistance

I-V graph for a filament lamp

As current flows through a filament lamp, the lamp heats up; resistance increases with temperature, causing the S-shaped curve

- As current through a filament lamp increases, the resistance **increases** because:
    - the higher current causes the **temperature** of the filament to increase
    - the higher temperature causes the atoms in the metal lattice of the filament to **vibrate** more
    - this causes an increase in resistance as it becomes more difficult for **free electrons** (the current) to pass through
    - since resistance **opposes** the current, this causes it to increase at a **slower** rate

I-V graph for a diode

- A diode allows current to flow in **one** direction only
    - This is called **forward bias**

- In the reverse direction, the diode has very **high resistance**, and therefore **no** current flows
    - This is called **reverse bias**

- When the current is in the direction of the arrowhead symbol, this is **forward bias**
    - On the IV graph, this is shown by a sharp increase in voltage and current on the right side of the graph
    - This shows the resistance is very **low**

- When the diode is switched around, this is **reverse bias**
    - On the IV graph, this is shown by a zero reading of current or voltage on the left side of the graph

The current is zero at all potential differences in the negative quadrants because current only flows one way through a diode; this gives the diode I-V graph its distinct shape

Examiner Tips and Tricks

In your IGCSE exam, you could be asked to recognise, sketch or explain the I-V graphs for a wire / fixed resistor (ohmic conductors), a filament lamp and a diode.

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